

Objectives:

- Compare different types databases
- Explain the difference between OLAP and OLTP environments
- Appraise various vendor databases

Activities:

1. What are the common features of all database systems? Identify at least five

Answer/Observation:

- *Support multiple users*
- *Less Redundancy and replication of data*
- *Delivers high level of security*
- *Saves Storage space and cost*
- *Data Storage management*
- *Backup and recovery management*

2. What are the unique features of each (i.e. any features that none of the other have)?

Answer/Observation:

- *Oracle*
 - *Has more security or advance security when purchase corporation edition (more powerful)*
 - *Supports OS such as Windows, Linux, Solaris, HP-UX ,OS X, z/OS , AIX, etc whereas SQL server only windows and Linux. (More OS support)*
 - *Database is shared among all schemas and users*
 - *Oracle supports star query optimization while SQL server doesn't offer query optimization (Query Optimization)*
 - *Allows rollback during the transaction process*
 - *Uses both "after" and "before" triggers*
 - *Offers Blockchain Tables.*
 - *Offers ACID transactional guarantee. In terms of CAP, it offers immediate Consistency as a single Server.*
- *SQL Server*

- *Displays error, and warning messages about the migration in a progress window*
- *An activity monitor feature with filtering and automatic refresh*
- *...*
- *PostgreSQL*
 - *Geo-Tagging*
 - *Define your own data types*
 - *write code from different programming languages without recompiling your database*
 - *It offers more Advanced indexes liked partial Index, Bloom Filters. Also, creating an Index in PostgreSQL is non-blocking.*

3. Comment on their suitability for OLAP vs OLTP workload

Answer/Observation:

- *Oracle: Supports both OLTP and OLAP workload. Best to use when traditional transactional workloads with structured (SQL) data and when ACID transaction guarantee is a key criterion. A feature offered by Oracle which Online Analytical Processing (OLAP) is suitable for the complex analysis of data, calculations which are useful in Organizations that would use these analytics to do forecasting. Whilst, Online Transaction Processing (OLTP) software is more suitable for real-time execution of large numbers of database transactions.*
- *SQL Server: If choose 'Analysis Services' as server type in SQL server management studio then its OLAP and if you choose 'Database Engine' then its OLTP. OLAP queries are mostly select operations while OLTP are insert, update and delete information from the database.*
- *PostgreSQL: Is a powerful database, and for OLAP workloads, it can certainly meet expectations. With a good deal of planning and tuning, the database engine will be able to deliver analytics at scale. PostgreSQL performs well in OLTP/OLAP systems when read/write speeds are required and extensive data analysis is needed.*

4. Compare the system specification requirements

	Oracle	SQL Server	PostgreSQL
Operating System Supported	<i>Windows, Linux, Solaris, HP-UX, OS X, z/OS, AIX, etc</i>	Windows & Linux	Windows, Mac and Linux
Hard drive space required	Minimum 500 MB free disk space for installation, 10 GB recommended	Minimum of 6 GB	512 MB
Memory required	Minimum 1 GB, 4 GB recommended	1 GB	2 GB
Processors Required	Intel Pentium® at 500 MHz or faster	2.0 GHz or faster	1 GHz

5. Do they have a cloud (database as a service) version available?

- Oracle – They provide a range of industry-leading on-premises and cloud-based solutions to meet the data management requirements from small, medium sized businesses to large global enterprises. Oracle Database Cloud Service provides pay-per-use or subscription-based access to Oracle databases, with pre-packaged bundles of options for application development, testing and production deployments. **YES**
- SQL Server – They offer SQL server in the cloud on Azure SQL with built-in security and manageability which can also migrate SQL workloads with SQL server on Azure virtual machines. It offers premium storage online. **YES**
- PostgreSQL -. PostgreSQL relational databases are available on Google Cloud Platform. Can be deployed on any cloud provider, with a variety of PostgreSQL hosting solutions available. PostgreSQL works in cloud architectures using ansible, kubernetes, and proprietary tools from multiple cloud vendors. **YES**

6. Compare the price for a comparative system setup (similar number of processors) for each database system.

A server with 4 CPUs and 4 cores per CPU would cost an estimate \$380,000 with Oracle, but just \$114,000 with SQL Server. The costs rise even higher with extra features such as table partitioning, data compression, and online analytical processing (OLAP). While there is no fee for PostgreSQL even for use in commercial software products.

7. Identify three (3) top companies that use each of these databases

Oracle

- ✓ Netflix
- ✓ LinkedIn
- ✓ eBay

SQL Server

- ✓ Stack Overflow
- ✓ Intuit
- ✓ Bank of America

PostgreSQL

- ✓ NVIDIA
- ✓ Skype
- ✓ Spotify

8. Which would you recommend for:

A. An OLTP for a multinational financial conglomerate with several TBs of data, thousands of users and dozens of locations worldwide

B. A small retail store wanting to manage their inventory

Justify your choice based on the findings of your investigation above.

I would recommend Oracle database for (A) reason being that the OLTP database software of choice should be able to perform necessary operations for the multinational financial conglomerate and Oracle would prove to be most suitable database system considering it is more powerful and have a wider range of features. With thousands of users and dozens of locations worldwide Oracle could even offer better identity and user controls along with its superior scalability since the company is dealing with a large storage of data.

For a small retail store wanting to manage their inventory I would recommend starting off with PostgreSQL which is an open source which is important for small startup and is free. Would not recommend Oracle as it would be too expensive and have features that may not even be utilize.